DS8001 Project Proposal: Pivot-Based Approximate Similarity Search for Hospital Encounters

Jared Smouter and Amarnadh Nalla | Project path: **Empirical Analysis** | Group of two

1. Problem and questions. Finding the most similar past hospital encounters supports case-based reasoning, cohort building and nearest-neighbor (kNN) risk prediction, but exact search computes a distance to every one of the N encounters. Following the scalable-indexing material of the course, we study the algorithmic question: *how well can pivot-based approximate indexes trade search quality against distance computations, index size and build cost on large, mixed-type clinical data?* (Q1) How do the number of pivots n, the pruned-list length ñ and the re-ranking budget c affect recall@K, and which dominates? (Q2) Does the metric permutation table (MPT) improve on the metric inverted file (MIF) accumulator, especially for small ñ? (Q3) How do the costs grow with N, and how does the intrinsic dimensionality of the data limit pruning, compared with exact tree indexes?

2. Algorithms and implementation. We will implement ourselves: (i) **exact brute-force** K-NN, as ground truth; (ii) **permutation-based indexing** (PBI), where each object is stored as the ordered list of its ñ nearest pivots; (iii) the **metric inverted file** (MIF) with its Spearman footrule accumulator and exact re-ranking of the best c candidates (Amato and Savino, 2008); and (iv) the **MPT** scoring refinement, plus a prefix index (Esuli, 2012) if time permits. We will also compare random against farthest-first pivot selection. The report will analyze build cost O(Nn) distances, index size O(Nñ), and why recall is limited when a query and its true neighbor share no pivot. Baselines (library code, not counted as our implementation): scikit-learn KD-tree and ball tree for exact search, and an inverted-file (IVF) index with k-means centroids (Jegou et al., 2011) built with scikit-learn.

3. Data. The **Diabetes 130-US Hospitals (1999–2008)** dataset (Strack et al., 2014; UCI Machine Learning Repository, CC BY 4.0): 101,766 inpatient encounters, 71,518 patients and 50 attributes (demographics, stay and utilization counts, ICD-9 diagnoses, 23 medication fields), with the 3-class label `readmitted` (NO, >30, <30; <30 is about 11%). Preprocessing: drop IDs and `weight` (about 97% missing), and keep missing lab results as an explicit category. Because pivot methods only need a metric, we can use a mixed-type distance (the mean of per-feature distances: range-normalized absolute difference for numeric and ordinal attributes, 0/1 mismatch for categorical ones, in the spirit of Gower, 1971), which is a proper metric. We will also test an all-numeric one-hot Euclidean encoding. Subsets of N = 10k, 25k, 50k and 100k provide scaling instances.

4. Experiments and metrics. On 1,000 held-out queries with K = 10 and 50 we report recall@K against brute-force ground truth. Following the lecture, we report machine-independent costs: distances computed per query, postings read per query, index size and build distances. Wall-clock query latency and peak memory are reported too, but only between methods run in the same language on the same machine. We run a grid over (n, ñ, c) and report the recall–cost trade-off curves for each method, then the scaling with N. Queries come from a patient-grouped split (by `patient_nbr`) so no patient appears in both the index and the queries. Pivot selection is random, so each setting is repeated over 10 seeds and reported as mean ± standard deviation. Parameters are tuned on a validation set of patients, not on the query set. As an optional extension, we test how approximate neighbors change kNN readmission prediction, using balanced accuracy and macro-F1.

5. Deliverables. A Python (3.11+) package with a README and fixed seeds, and a report connecting the empirical trade-offs to the theory of pivot-based indexing and the curse of dimensionality.

References

Amato, G., and Savino, P. (2008). Approximate similarity search in metric spaces using inverted files. In *Proc. 3rd International Conference on Scalable Information Systems (InfoScale)*.

Esuli, A. (2012). Use of permutation prefixes for efficient and scalable approximate similarity search. *Information Processing & Management*, 48(5), 889–902.

Gower, J. C. (1971). A general coefficient of similarity and some of its properties. *Biometrics*, 27(4), 857–871.

Jegou, H., Douze, M., and Schmid, C. (2011). Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(1), 117–128.

Mohamed, H., and Marchand-Maillet, S. (2015). Scalable indexing for big data processing. In *Big Data: Algorithms, Analytics, and Applications*. Chapman and Hall/CRC.

Strack, B., DeShazo, J. P., Gennings, C., Olmo, J. L., Ventura, S., Cios, K. J., and Clore, J. N. (2014). Impact of HbA1c measurement on hospital readmission rates: analysis of 70,000 clinical database patient records. *BioMed Research International*, 2014, Article 781670. Data: UCI Machine Learning Repository, doi:10.24432/C5230J.